Calibration results
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Normalized Residuals

Reprojection error (cam0): mean 0.1999724783464811, median 0.1931678679363629, std: 0.1046879869029291
Reprojection error (cam1): mean 0.197178805224744, median 0.18426069305270915, std: 0.10492954688091712
Gyroscope error (imu0): mean 0.48141352183170216, median 0.3957213502684498, std: 0.5181623110181255
Accelerometer error (imu0): mean 0.45878876235735727, median 0.35634854649062647, std: 0.3570585170122871

Residuals

Reprojection error (cam0) [px]: mean 0.1999724783464811, median 0.1931678679363629, std: 0.1046879869029291
Reprojection error (cam1) [px]: mean 0.197178805224744, median 0.18426069305270915, std: 0.10492954688091712
Gyroscope error (imu0) [rad/s]: mean 0.006233857252174818, median 0.005124223349243154, std: 0.006709719885004054
Accelerometer error (imu0) [m/s^2]: mean 0.027804703827087877, median 0.02159635677097843, std: 0.021639384241789406

Transformation (cam0):

T_ci: (imu0 to cam0):
[[0.99986917 -0.0112869   0.01158662 0.00579037]
 [0.01138189 0.99990189 -0.00816482 -0.00513003]
 [-0.01149333 0.00829562 0.99989954 -0.01387312]
 [0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[0.99986917 0.01138189 -0.01149333 -0.00589068]
 [-0.0112869   0.99990189 0.00829562 0.00530997]
 [0.01158662 -0.00816482 0.99989954 0.01376275]
 [0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.004981929251041623

Transformation (cam1):

T_ci: (imu0 to cam1):
[[0.99986917 -0.0112869   0.01158662 -0.04451659]
 [0.01138189 0.99990189 -0.00816482 -0.00513003]
 [-0.01149333 0.00829562 0.99989954 -0.01387312]
 [0.   0.   0.   1.   ]]

T_ic: (cam1 to imu0):
[[0.99986917 0.01138189 -0.01149333 0.04440971]
 [-0.0112869   0.99990189 0.00829562 0.00474216]
 [0.01158662 -0.00816482 0.99989954 0.01434564]
 [0.   0.   0.   1.   ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.0049723419861791044

Baselines:

Baseline (cam0 to cam1):
[[1.   0.   0.   -0.05030697]
 [0.   1.   0.   0.   ]
 [0.   0.   1.   0.   ]
 [0.   0.   0.   1.   ]]
baseline norm: 0.05030696839094162 [m]

Gravity vector in target coords: [m/s^2]
[-0.12373393 -9.80545504 0.0785127]

Calibration configuration
=========================

cam0

Camera model: pinhole
Focal length: [432.3992004394531, 432.3992004394531]
Principal point: [421.42779541015625, 238.08770751953125]
Distortion model: radtan
Distortion coefficients: [0.0, 0.0, 0.0, 0.0]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.055 [m]
Spacing 0.0165 [m]

cam1

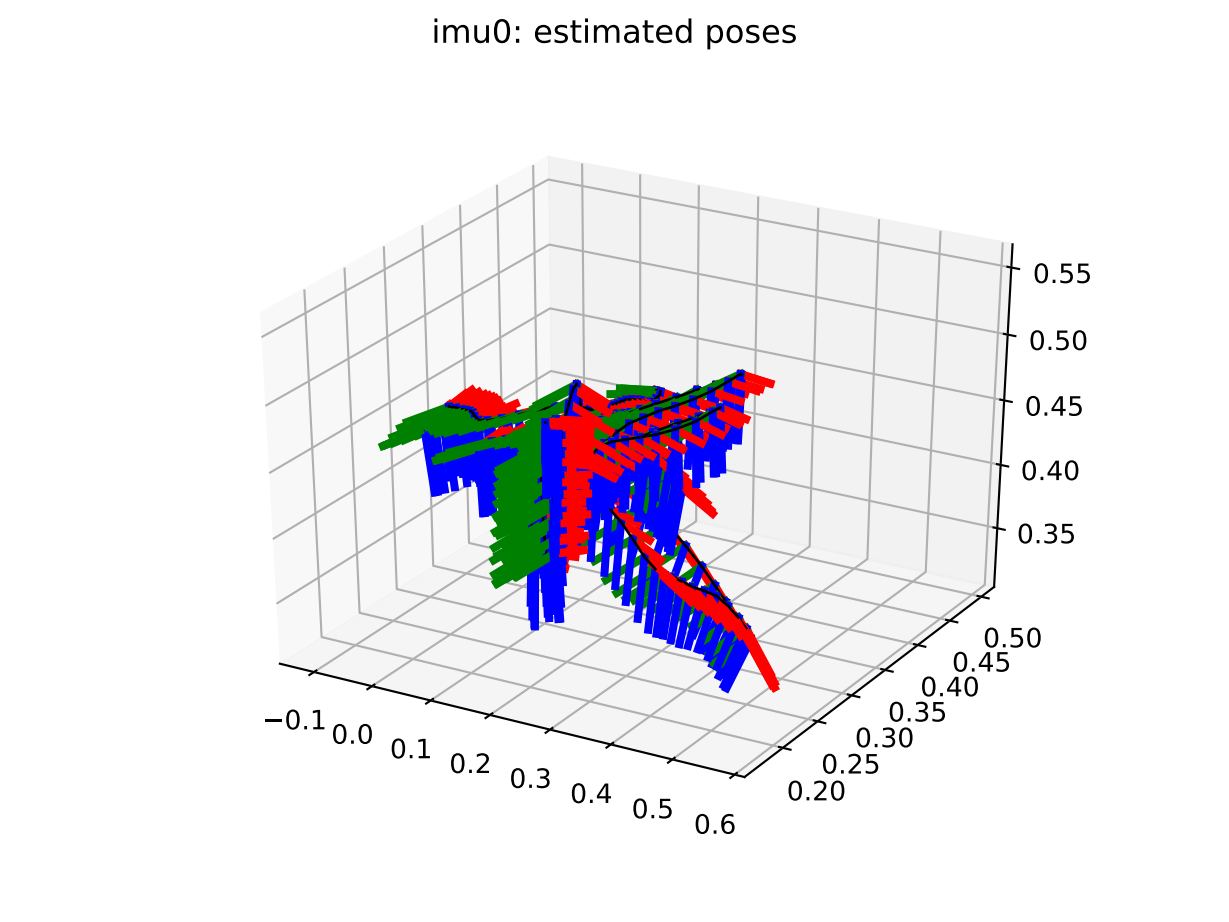
Camera model: pinhole
Focal length: [432.3992004394531, 432.3992004394531]
Principal point: [421.42779541015625, 238.08770751953125]
Distortion model: radtan
Distortion coefficients: [0.0, 0.0, 0.0, 0.0]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.055 [m]
Spacing 0.0165 [m]

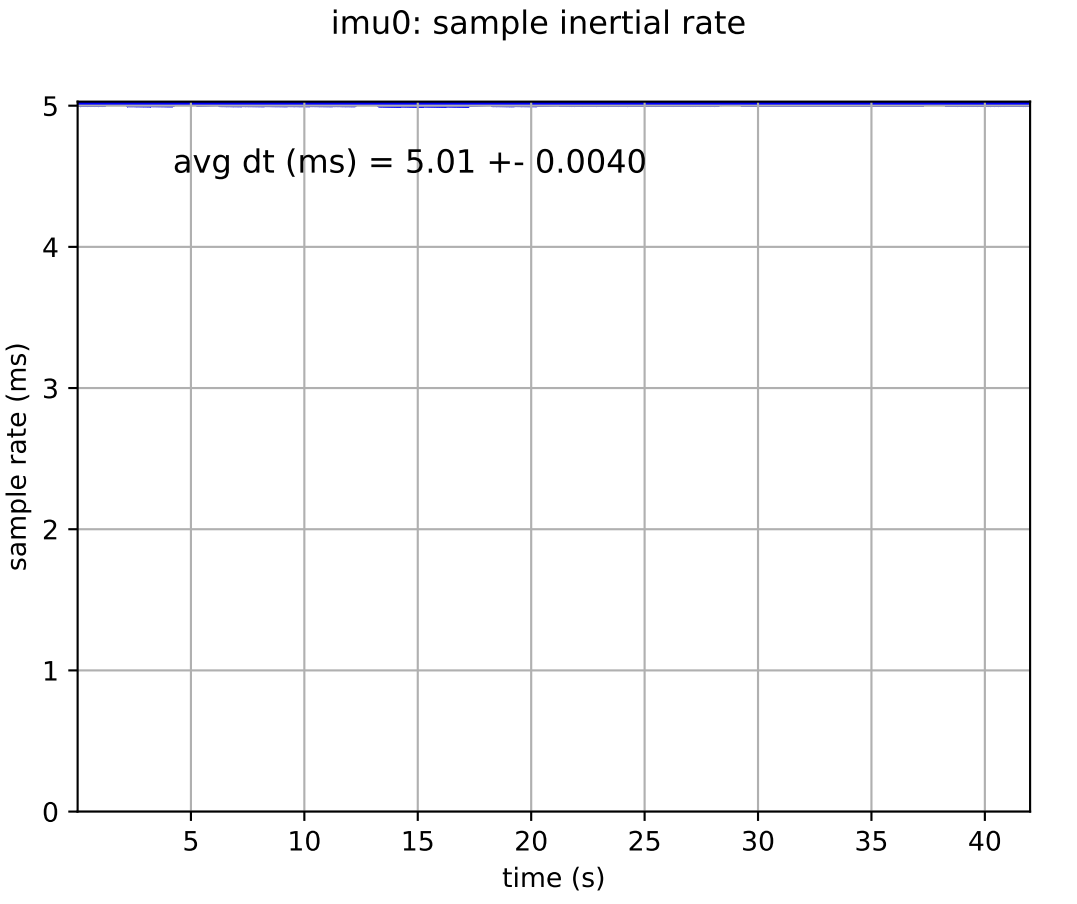
IMU configuration
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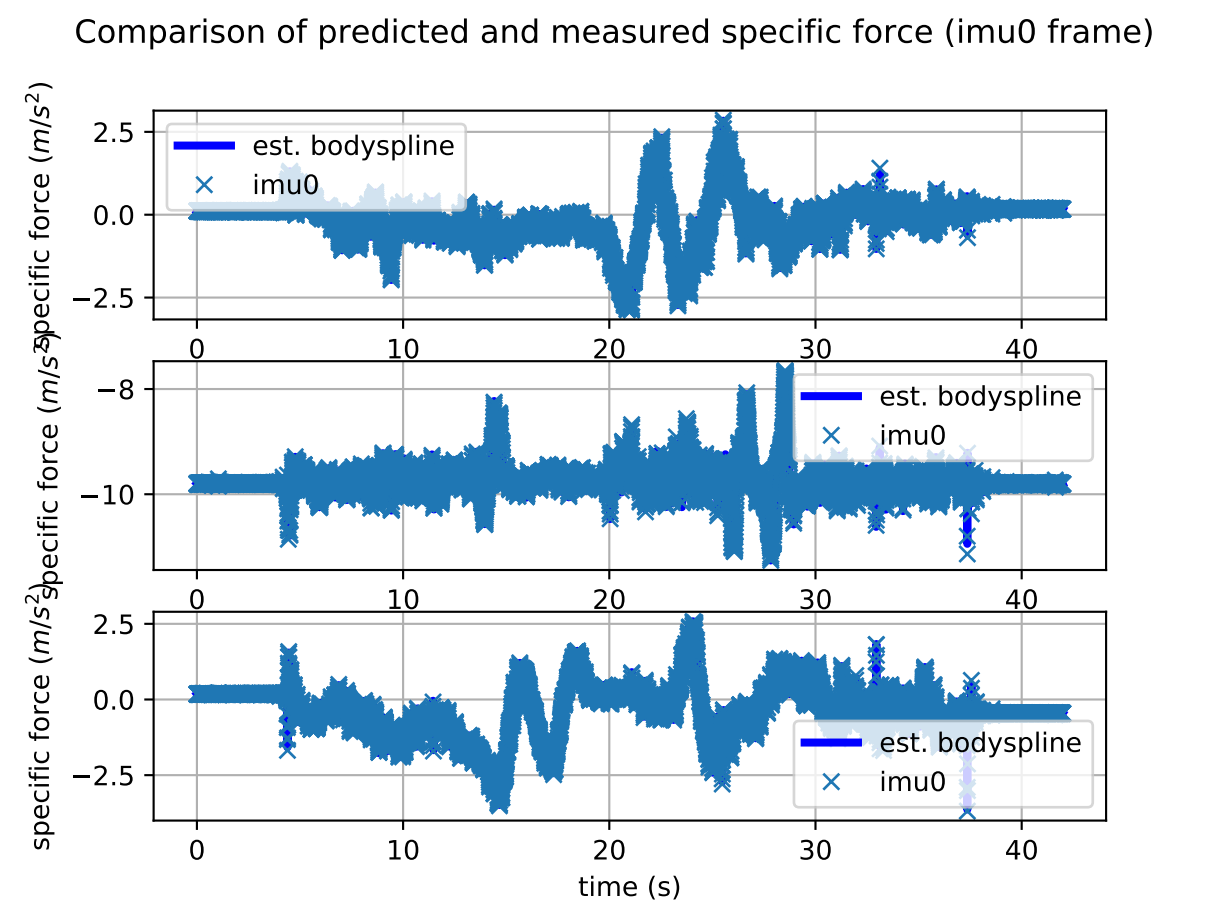
IMU0:

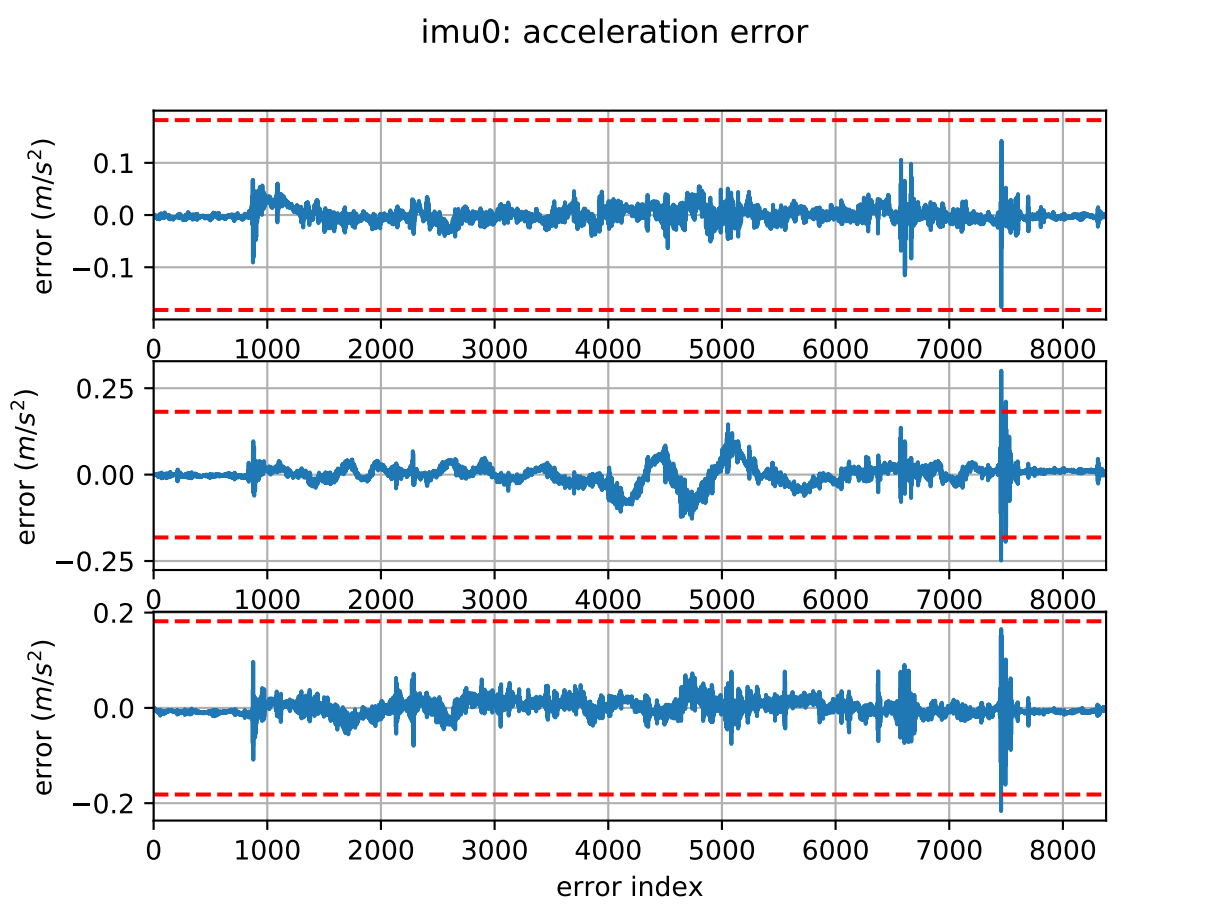
Model: calibrated
Update rate: 200.0

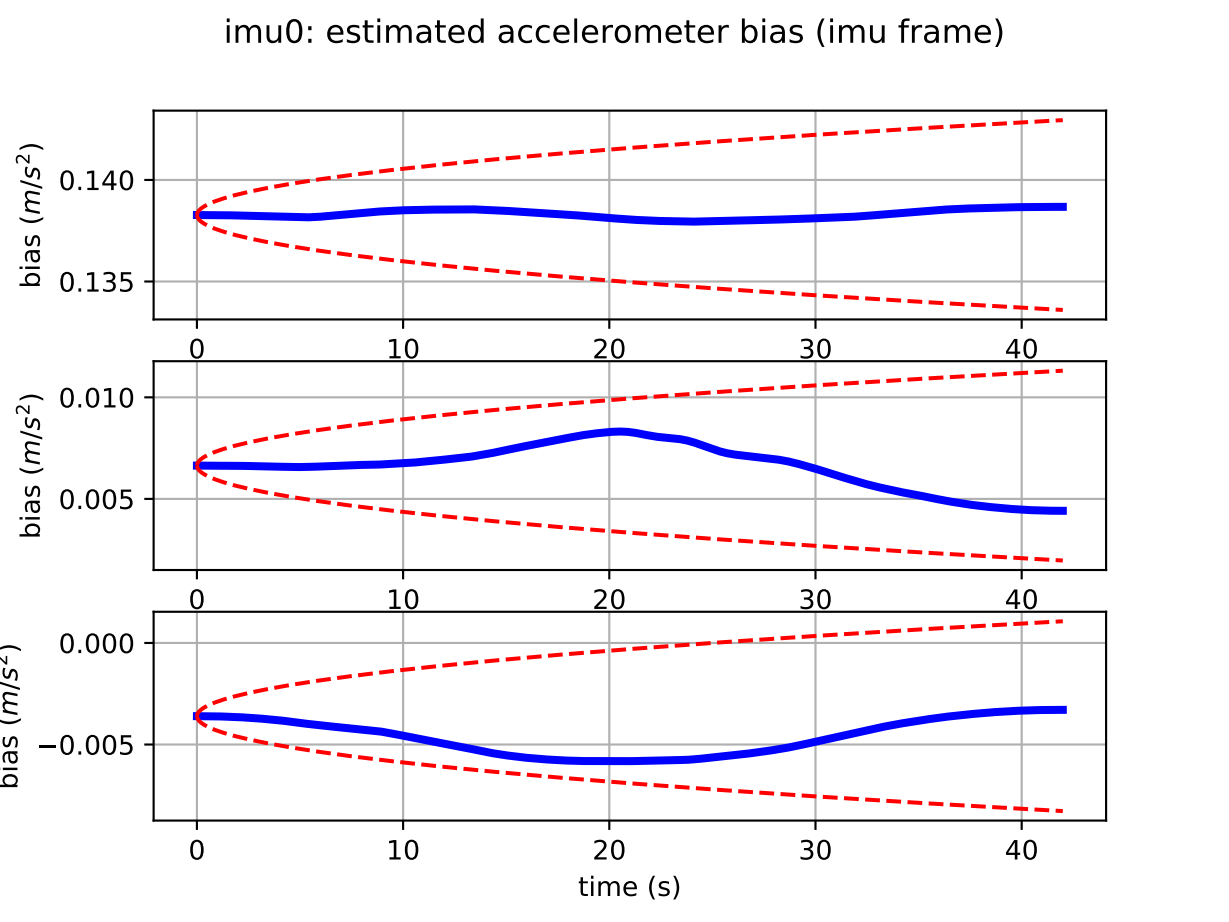
Accelerometer:
Noise density: 0.0042853915
Noise density (discrete): 0.06060458779378382
Random walk: 0.0002403472
Gyroscope:
Noise density: 0.0009156375
Noise density (discrete): 0.012949069707173948
Random walk: 2.158951e-05
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

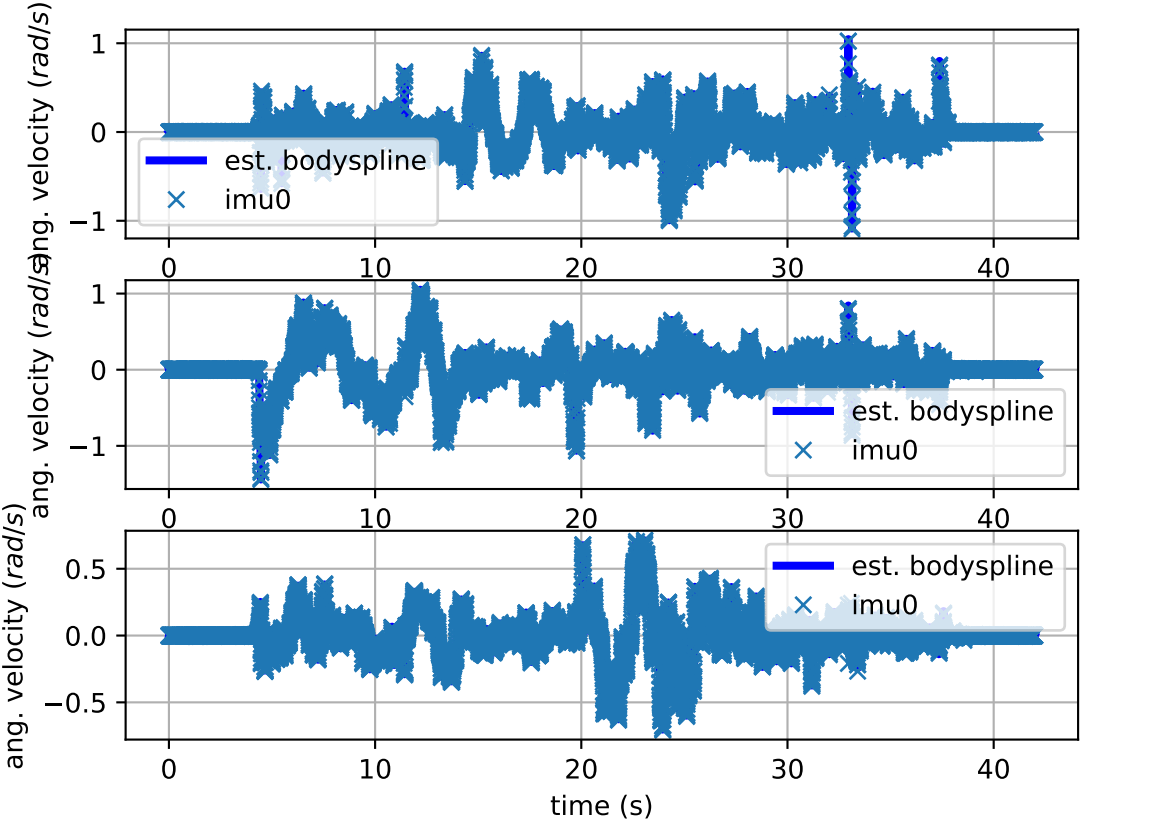

imu0: estimated poses
0.55
0.50
0.45
0.40
0.35
0.50
0.45
0.40
0.35
0.30
0.25
0.20
−0.1
0.0
0.1
0.2
0.3
0.4
0.5
0.6

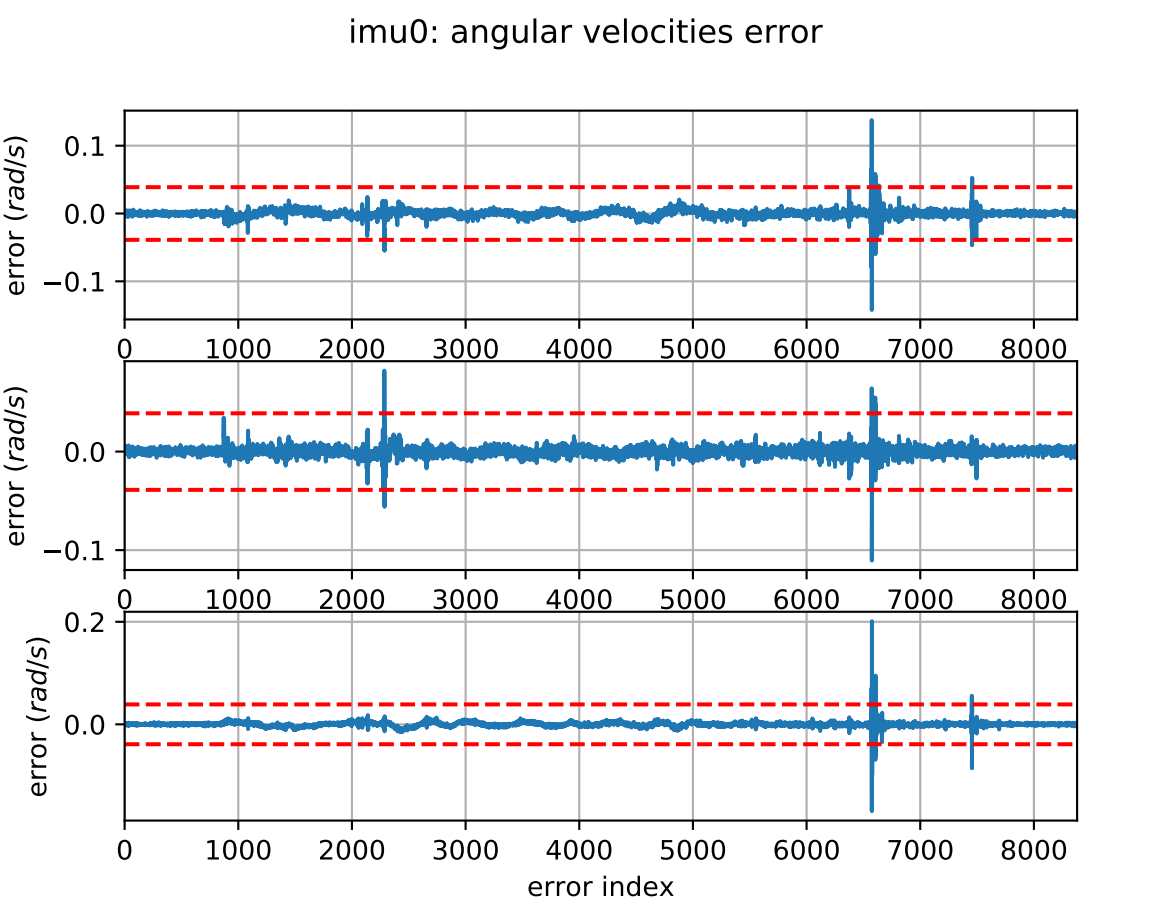

imu0: sample inertial rate
avg dt (ms) = 5.01 +- 0.0040
sample rate (ms)
time (s)
0
1
2
3
4
5
5
10
15
20
25
30
35
40

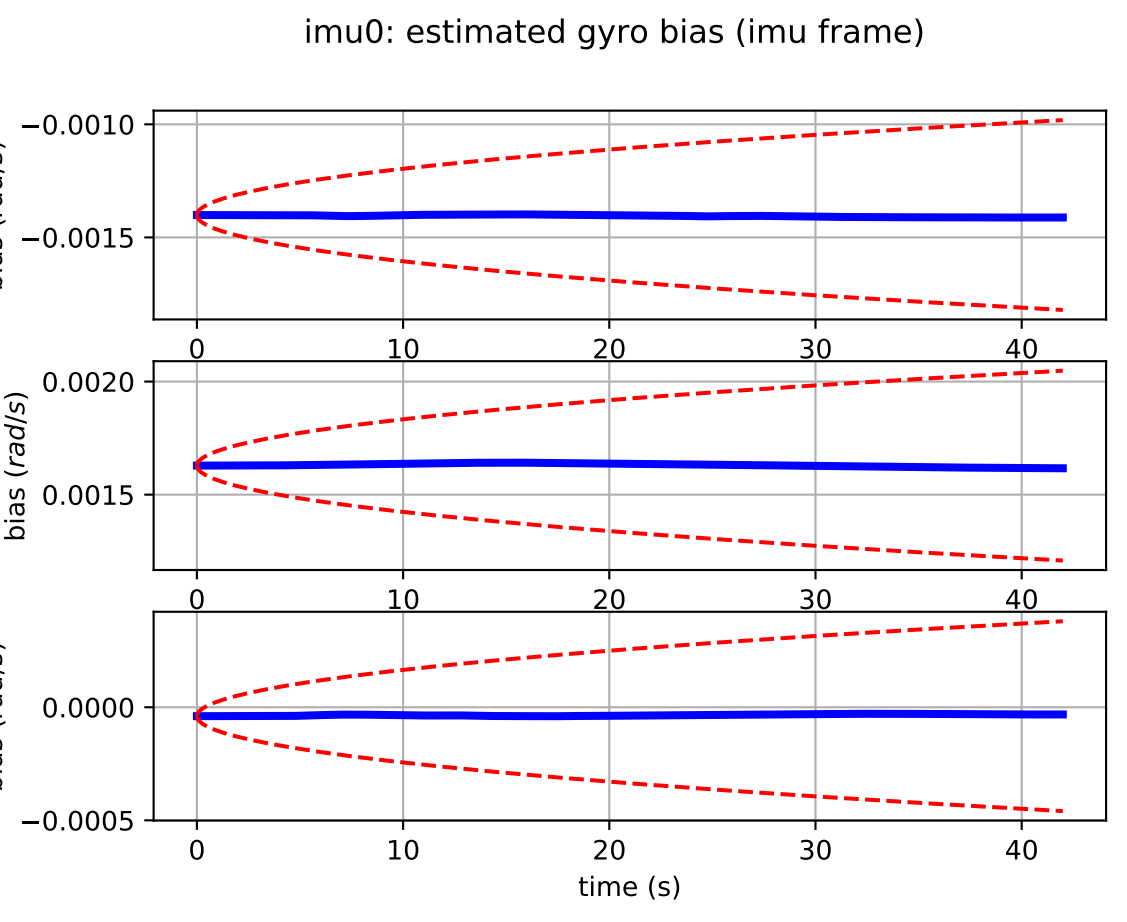

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
2.5
0.0
−2.5
−8
−10
0
10
20
30
40
time (s)

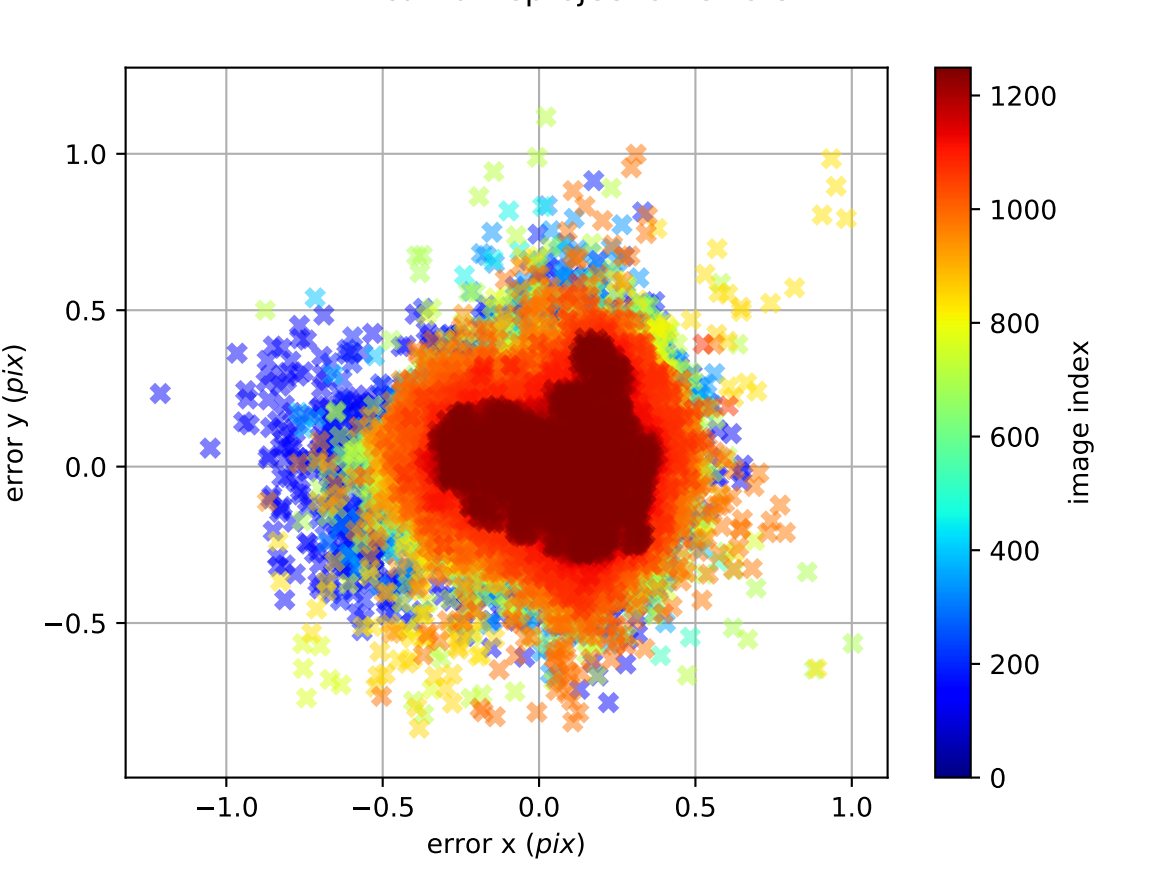

imu0: acceleration error
error (m/s^2)
0.1
0.0
−0.1
0
1000
2000
3000
4000
5000
6000
7000
8000
error (m/s^2)
0.25
0.00
−0.25
0
1000
2000
3000
4000
5000
6000
7000
8000
error (m/s^2)
0.2
0.0
−0.2
0
1000
2000
3000
4000
5000
6000
7000
8000
error index

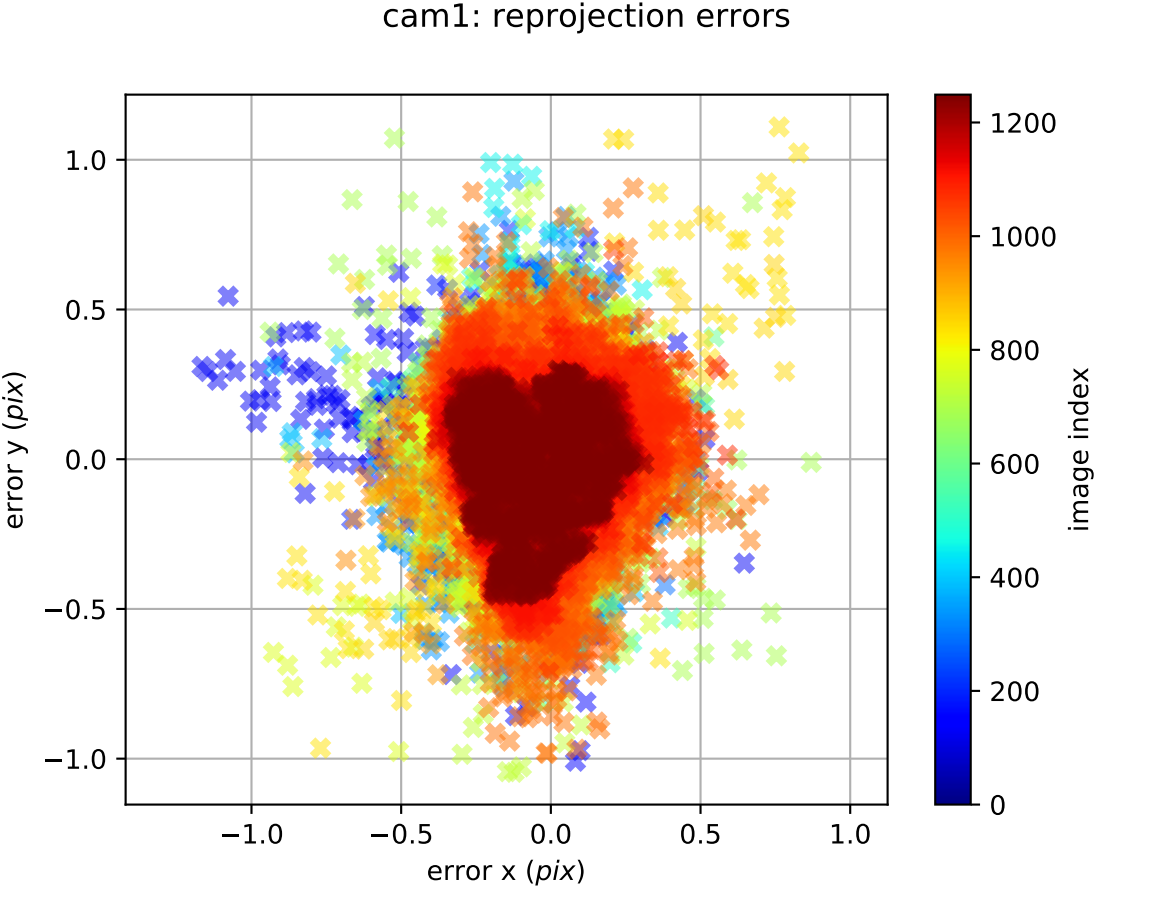

imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.140
0.135
0
10
20
30
40
bias (m/s^2)
0.010
0.005
0
10
20
30
40
bias (m/s^2)
0.000
−0.005
0
10
20
30
40
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
ang. velocity (rad/s)
ang. velocity (rad/s)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
time (s)


imu0: angular velocities error
error (rad/s)
0.1
0.0
−0.1
0
1000
2000
3000
4000
5000
6000
7000
8000
error (rad/s)
0.0
−0.1
0
1000
2000
3000
4000
5000
6000
7000
8000
error (rad/s)
0.2
0.0
0
1000
2000
3000
4000
5000
6000
7000
8000
error index


imu0: estimated gyro bias (imu frame)
−0.0010
−0.0015
0.0020
0.0015
0.0000
−0.0005
bias (rad/s)
0
10
20
30
40
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
1.0
0.5
0.0
−0.5
−1.0
1200
1000
800
600
400
200
0


cam1: reprojection errors
error x (pix)
error y (pix)
image index
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
1200
1000
800
600
400
200
0